DATA MINING & MACHINE LEARNING

Classification of Human Body Postures through Wearable Sensors

Data Mining & Machine Learning Project
M.Sc. in Artificial Intelligence and Data Engineering — Academic Year 2025–2026

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OVERVIEW

Presentation Outline

01 **Problem & Dataset**
Task definition · features · EDA

02 **Pipeline & Experiments**
Preprocessing · model selection · hyperparameter optimization

PART 1

03 **Evaluation & Results**
Metrics · cross-validation · final model performance

04 **Explainability**
Global & local techniques · consistency with domain knowledge

PART 2

05 **Critical Discussion**
Error analysis · limitations · future directions

PART 1 — TECHNICAL

Problem, Pipeline & Results

Problem definition · feature engineering · model selection
hyperparameter optimization · evaluation · test-set results

PROBLEM

Problem Definition & Dataset

The Problem

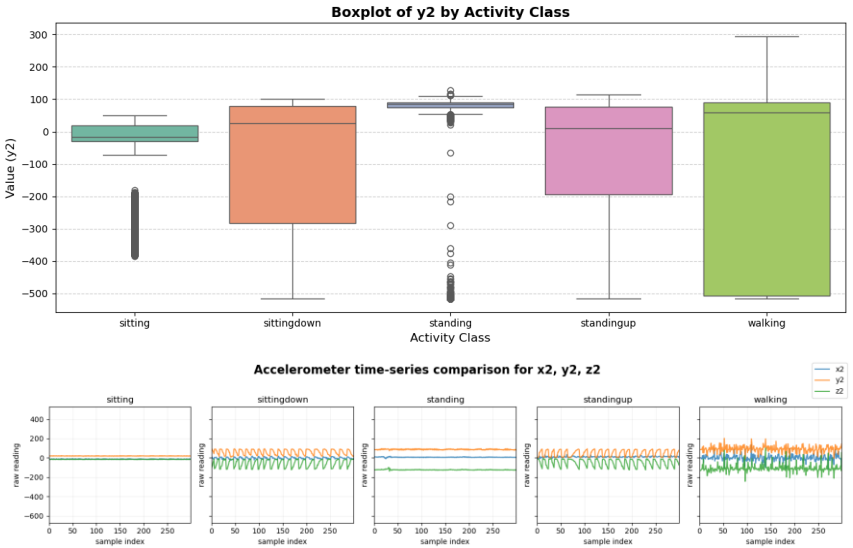
- Human Activity Recognition (HAR) system using 4 tri-axial wearable accelerometers positioned on the body.
- Task type: Multi-class classification (5 target postures and movements).
- Ambient assisted living, elderly remote care, physical rehabilitation.
- High inter-subject variability using only 4 subjects and low-frequency sensors (8 Hz).

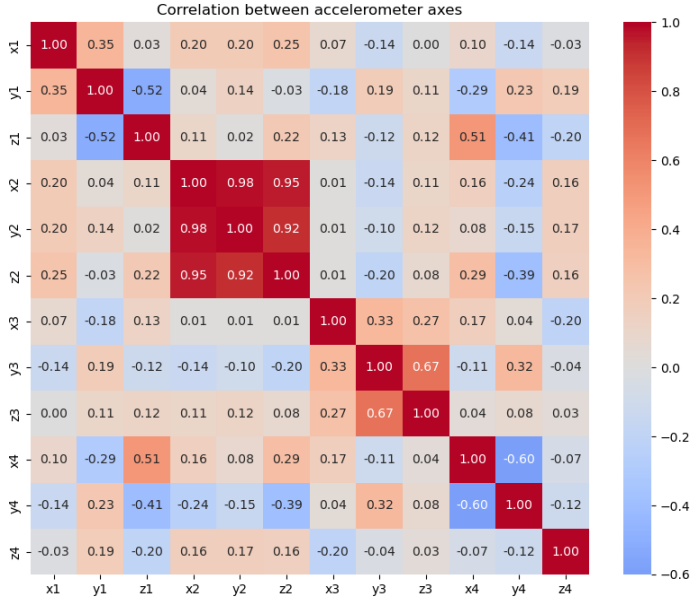
Dataset at a Glance

Task	*Multi-class classification*
Dataset	*UCI Wearable Computing: Classification of Body Postures and Movements (PUC-Rio) --- 165,632 samples, 18 features*
Target	*class (sittingdown, standingup, standing, sitting, walking)*
Features	*18 features: 16 numerical, 2 categorical*
Missing values	*No missing values, one malformed cell*
Class balance	*Sitting standing walking standingup sittingdown 30.57% 28.60% 26.20% 7.50% 7.14%*
Split	*Train/Val/Test Held-out user: 3 users%/3-Fold/1 user*
Metric(s)	*Weighted F1-Score, ROC-AUC*

PROBLEM

Exploratory Data Analysis



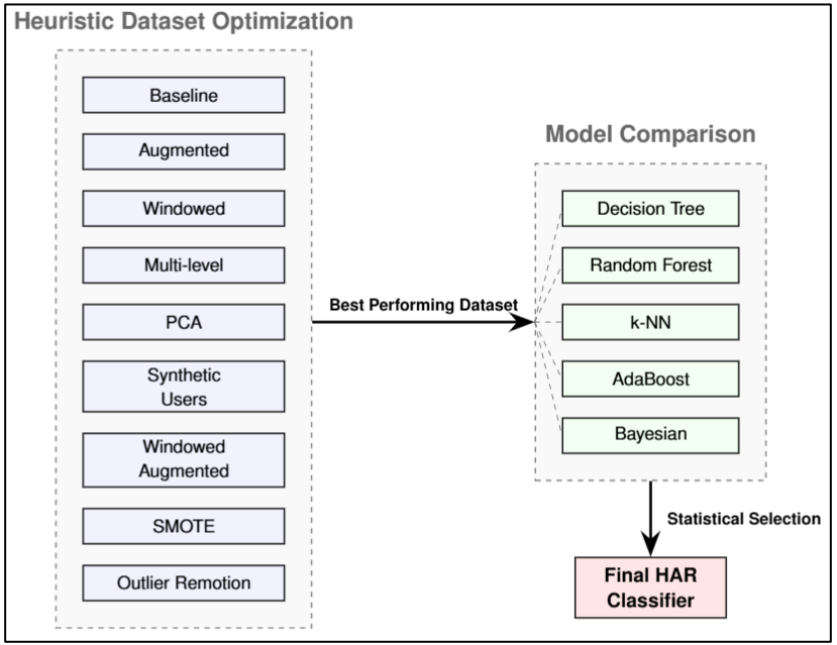


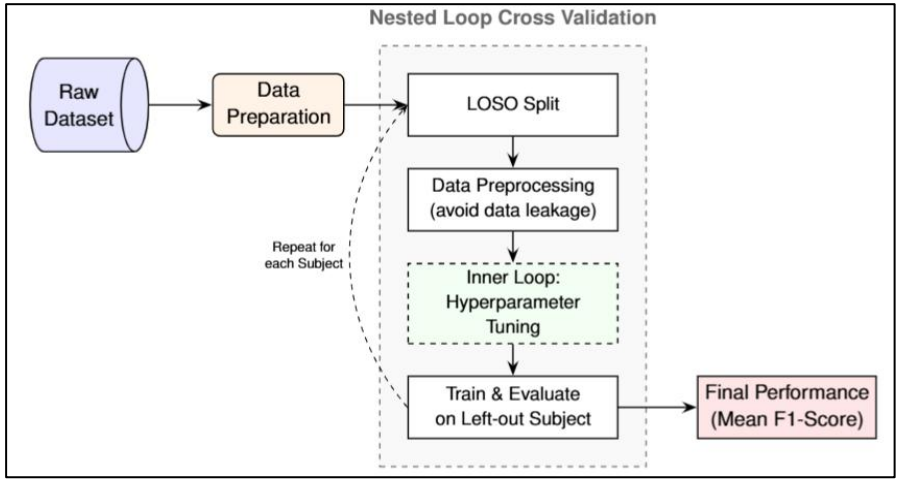
Key findings from EDA:

1. Imbalances in classes
2. Dynamic activities (walking) show significantly larger variance and peak amplitudes compared to static positions (sitting, standing).
3. Sensors placed on the left thigh (Sensor 2) and right ankle (Sensor 3) show the highest spatial separability of physical movements.
4. Linear correlation is present exclusively within axes of the same sensor

PIPELINE

Overall and ML pipelines





PIPELINE

Feature Engineering & Selection

Engineering

- Magnitude, Roll, Pitch, and Yaw for each tri-axial accelerometer: whole dataset.
- Rolling windows of 8 samples and statistical metrics: whole dataset.
- Z-score standardization was applied for the K-NN method: train set.
- Equal width binning discretization has been tested: train set.
- Categorical nominal variables (gender) were encoded using One-Hot Encoding: whole dataset.
- No missing values, only one single malformed row.

Selection Method

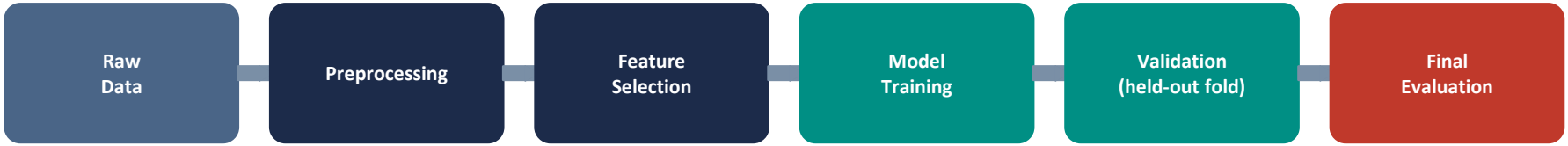
- Embedded feature selection: PCA on training set.
- Wrapper: feature-importance via global SHAP values.
- Criterion: Average absolute SHAP value (mean |SHAP|).
- Validated by comparing the average outer Leave-One-Subject-Out (LOSO) cross-validated weighted F1-score of different dataset configurations.
- Final features: 48 statistical features based on temporal windows

Why These Features?

- Window-based statistics (mean, variance) capture the short-term dynamics of movement while smoothing out raw, high-frequency signal noise.
- Age, weight, height, and BMI were discarded because they caused subject-specific overfitting.

PIPELINE

ML Pipeline: Avoiding Data Leakage & Bias



Data Leakage Prevention

Each experiment's preprocessing is learnt on the training set only:

- PCA
- Discretization
- SMOTE
- Outlier remotion

Class Imbalance Handling

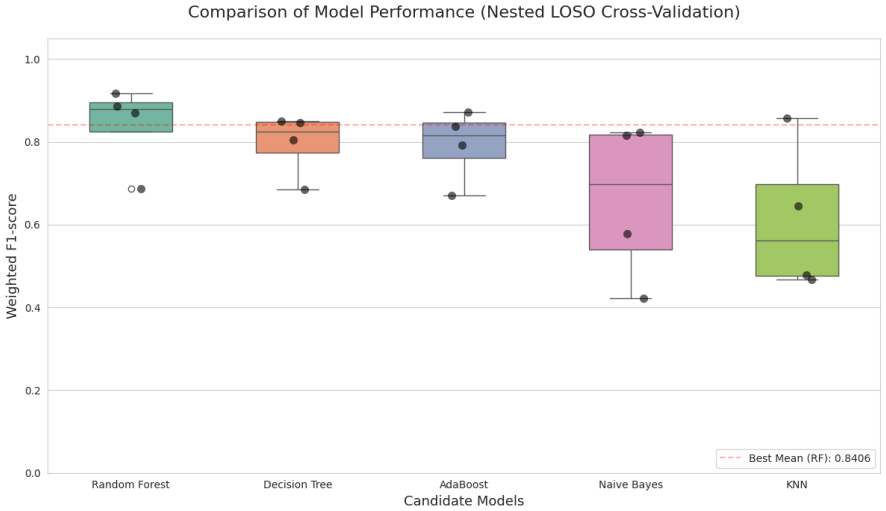
- Strategy tested: SMOTE
- Applied inside CV folds only, given the application of LOSO in the outlier folds
- Weighted F1-score and multi-class ROC-AUC curves were selected

Metric Justification

- Weighted F1-score is highly suitable as penalizes errors coherently with classes frequency.
- A False Negative could leave a patient's transition unmonitored or miss a fall, whereas a False Positive overestimates user mobility and causes alert fatigue for clinical staff.
- Aggregated confusion matrix and class-specific ROC-AUC curves.

PIPELINE

Model Selection & Hyperparameter Optimization

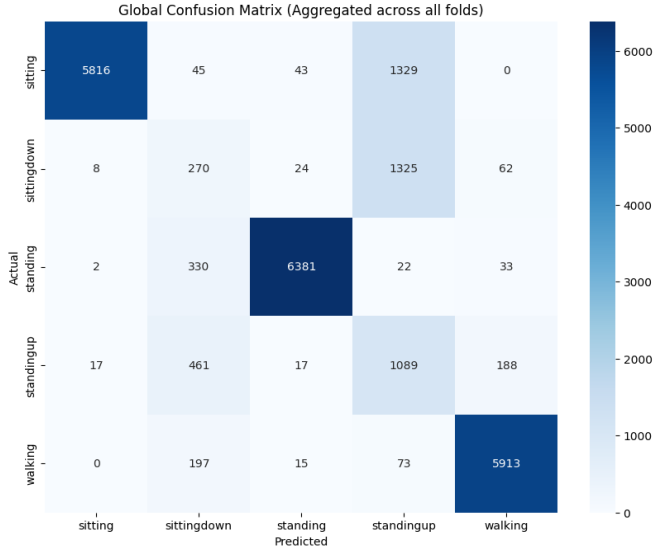


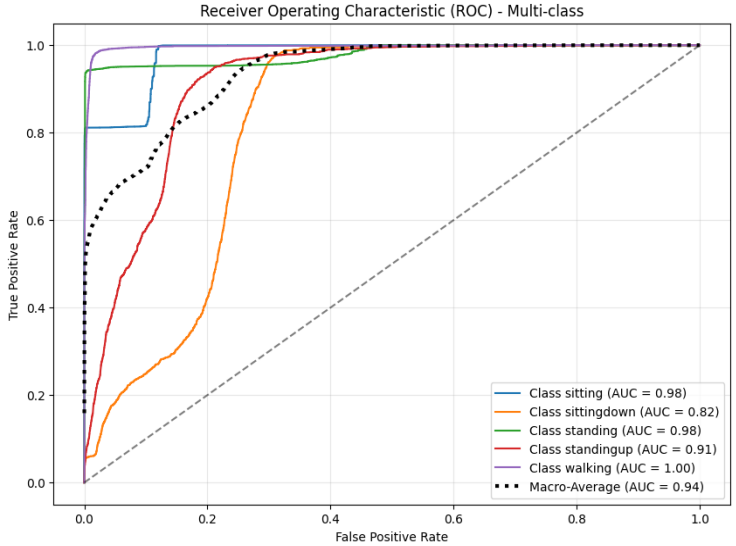
Key Decisions

Candidate models	*Decision Tree, Random Forest, AdaBoost, KNN, Gaussian Naïve Bayes*
CV strategy	*LOSO Nested CV*
Selection criterion	*Weighted F1-Score*
Chosen model	*Random Forest*
Search strategy	*Grid Search for KNN – Random Search others*
Search space	Random Forest: n_estimators: {100,200,300,500}; max_depth: {None,10,20,30,50}; min_samples_split: {2,5,10}; min_samples_leaf: {1,2,4}; max_features: {sqrt,log2}; bootstrap: {True,False}
Best config	Random Forest: n_estimators: {500}; max_depth: {10}; min_samples_split: {10}; min_samples_leaf: {1}; max_features: {log2}; bootstrap: {True}
Statistical test	*Wilcoxon test and t-test adaptation*

RESULTS

Test-Set Results





Key Metrics (Test Set)

0.83
Outer Weighted F1-score

0.93
Inner Weighted F1-score

0.84
Average AUC

0.3783
Baseline Weighted F1-score

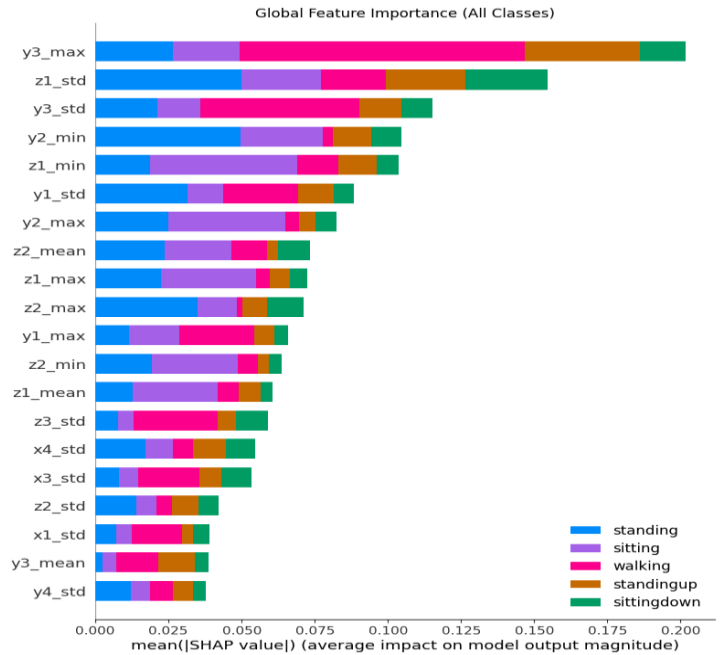
PART 2 — CRITICAL ANALYSIS

Explainability & Critical Discussion

How does the model work? · What did it get wrong and why?
Limitations · future directions · domain impact

EXPLAINABILITY

Global Explainability: What Drives the Model Overall?

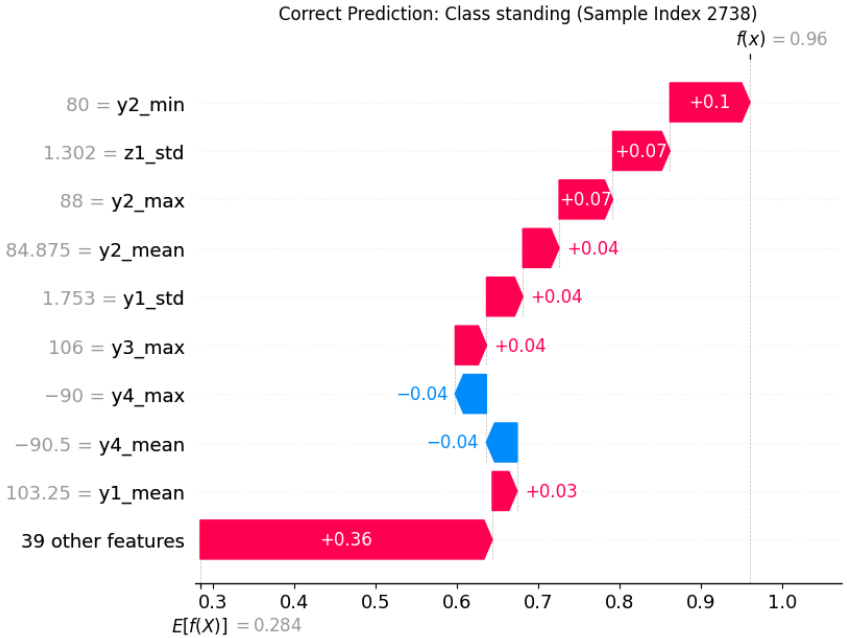


Interpretation

- Global SHAP feature importance.
- Most important features: Standard deviation and mean statistics on lower limbs. This matches domain expectations, as the thigh (sensor 2) and ankle (sensor 3) undergo the largest spatial and tilt rotations.
- Sensor 1 (on waist) turned out to be more important than what initial data exploration revealed. This is reasonable given its location.
- Sensor 4 (on right upper arm) confirms expectations: it doesn't allow to capture differences in positions
- Global explanation reveals that the classifier does not over-rely on a single sensor but uses a distributed network of bodily motion cues.

EXPLAINABILITY

Local Explainability: Why This Specific Prediction?

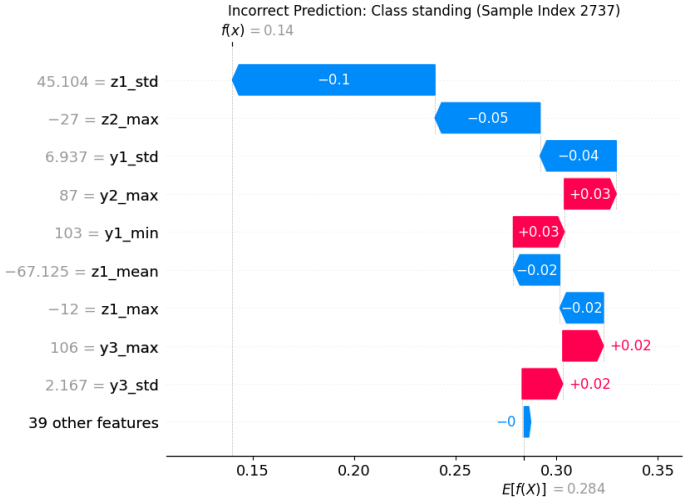


Case Analysis

- Representative correctly classified "standing" sample
- For the correct classification, the combination of global most important features contributed, coherently with domain knowledge
- Sensor 4 (upper-arm) is confirmed to be misleading, coherently with domain knowledge and global importance ranking.

CRITICAL DISCUSSION

Error Analysis: What Did the Model Get Wrong and Why?



Reverse-Engineering Errors

- Tipycal error pattern: sittingdown or standingup involved.
- Standing sample has been confused with standingup.
- Most features values contribution is around zero: no clear activity distinction.
- Coherently with domain knowledge, two similar positions are confused.
- Curiosity: minority class has been predicted over majority class. Model is confusing positions, and few values are sufficient to influence predictions.
- Usually, 39 other features have high contribution, here it is 0, which confirms model confusion.

Comparison with Prior Work

Original paper (Ugulino et al., 2012) claimed 99% accuracy using standard k-fold CV. However, k-fold suffers from severe subject-level data leakage. Our LOSO F1-score of 0.83 is lower but represents a realistic and honest performance estimate for unseen subjects.

CRITICAL DISCUSSION

Domain Impact, Limitations & Future Directions

Domain Impact

- This system can be integrated into healthcare or medical wearables to continuously monitor fail or elderly patients.
- Support for quantitative mobility mapping for clinical diagnostics and real-time triggers for medical emergency warnings (e.g., unexpected falls).
- False Negatives are critical as they fail to flag important activity changes; False Positives cause alarm fatigue and over-rely on patient independence.
- Designed as a clinical decision-support tool; medical expert oversight is required, as wearable data should complement physical exams, not replace them.

Limitations

- Small dataset (only 4 participants), not representative of particular subjects. This affects generalization capabilities and statistical tests.
- Imbalanced dataset for transitional classes
- Lack of long-term sequential temporal context across subsequent window blocks.
- Evaluated on a single, laboratory-controlled dataset; no field deployment testing under realistic, unstructured environments.
- invasive sensors don’t allow continual monitoring
- Not useful for Smartwatches since Sensor 4 (upper arm) resulted not important one

Future Directions

- Expand data collection to include 30+ participants across diverse age groups and clinical physical conditions.
- Implement transition logic rules based on the previous state (e.g., if sitting, the user cannot physically perform "sittingdown" next).
- Leverage sequential architectures like Recurrent Neural Networks to capture temporal transitions.
- Federated Learning

Thank You

Conclusions:

- Automated multi-class posture and movement classification using low-frequency (8 Hz) accelerometers.
- An optimized Random Forest classifier (F1-score: 0.83) validated through Leave-One-Subject-Out (LOSO) nested cross-validation.
- Transitional movements are confused without the context in which they happen.
- The primary limit is subject-dependence from having only 4 users.

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